

2019 Year 6 H3 Math Prelim Exam Solutions

Qn	Solution	Remarks
1(i)	$\frac{dx}{dt} = \frac{(c+t)x}{1-t^2}$ $\frac{1}{x} \frac{dx}{dt} = \frac{(c+t)}{1-t^2}$ $\int \frac{1}{x} dx = \int \frac{c}{1-t^2} - \frac{1}{2} \frac{-2t}{1-t^2} dt$ $\ln x - d = \frac{c}{2} \ln \left \frac{1+t}{1-t} \right - \frac{1}{2} \ln 1-t^2 $ $\ln \left \frac{x}{k} \right = \ln \left \frac{1+t}{1-t} \right ^{\frac{c}{2}} - \ln (1+t)(1-t) ^{\frac{1}{2}}$ $\left \frac{x}{k} \right = \frac{ 1+t ^{\frac{c-1}{2}}}{ 1-t ^{\frac{c+1}{2}}}$ $x = \frac{p 1+t ^{\frac{c-1}{2}}}{ 1-t ^{\frac{c+1}{2}}}, \text{ where } p \text{ is an arbitrary constant.}$	
(ii)	$y = tx$ $\frac{dy}{dx} = t + x \frac{dt}{dx}$ $t + x \frac{dt}{dx} = \frac{ax + b(tx)}{bx + a(tx)}$ $t + x \frac{dt}{dx} = \frac{a + b(t)}{b + at}$ $x \frac{dt}{dx} = \frac{a + b(t)}{b + at} - t$ $x \frac{dt}{dx} = \frac{a - at^2}{b + at}$ $\frac{1}{x} \frac{dx}{dt} = \frac{b + at}{a(1-t^2)}$ $\frac{dx}{dt} = \left(\frac{b}{a} + t \right) x$ and the resulting equation is equivalent to equation (A) with $c = \frac{b}{a}$.	

1(iii)	$\frac{ x }{ k } = \frac{\left 1 + \frac{y}{x}\right ^{\frac{b-1}{2}}}{\left 1 - \frac{y}{x}\right ^{\frac{b+1}{2}}}$ $ x x-y ^{\frac{b+1}{2}} x ^{-\frac{b+1}{2}} = k x+y ^{\frac{b-1}{2}} x ^{-\frac{b-1}{2}}$ $ x x-y ^{\frac{b+1}{2}} = k x+y ^{\frac{b-1}{2}} x ^{\frac{b+1}{2} - \frac{b-1}{2}}$ $ x x-y ^{\frac{b+a}{2a}} = k x+y ^{\frac{b-a}{2a}} x $ $ x x-y ^{b+a} = k ^{2a} x+y ^{b-a} x $ $ x-y ^{b+a} = D x+y ^{b-a}$	
		Total Marks: 11

Qn	Solution	Remarks
2(i)	Using the substitution $y = a + b - x$, we have $\begin{aligned}\int_a^b f(x) \, dx &= \int_b^a -f(a + b - y) \, dy \\ &= \int_a^b f(a + b - y) \, dy \\ &= \int_a^b f(a + b - x) \, dx \quad (\text{shown})\end{aligned}$	
(ii)	$\begin{aligned}&\int_0^{\frac{\pi}{4}} \ln(1 + \tan \theta) \, d\theta \\ &= \int_0^{\frac{\pi}{4}} \ln\left(1 + \tan\left(\frac{\pi}{4} - \theta\right)\right) \, d\theta \\ &= \int_0^{\frac{\pi}{4}} \ln\left(1 + \frac{\tan\left(\frac{\pi}{4}\right) - \tan \theta}{1 + \tan\left(\frac{\pi}{4}\right)\tan \theta}\right) \, d\theta \\ &= \int_0^{\frac{\pi}{4}} \ln\left(1 + \frac{1 - \tan \theta}{1 + \tan \theta}\right) \, d\theta \\ &= \int_0^{\frac{\pi}{4}} \ln\left(\frac{2}{1 + \tan \theta}\right) \, d\theta \quad (\text{shown})\end{aligned}$	
(iii)	$\int_0^1 \frac{\ln(1+x)}{1+x^2} \, dx = \int_0^{\frac{\pi}{4}} \ln(1 + \tan \theta) \, d\theta$ by using the substitution $x = \tan \theta$. Also, we have $\int_0^1 \frac{\ln(1+x)}{1+x^2} \, dx = \int_0^{\frac{\pi}{4}} \ln\left(\frac{2}{1 + \tan \theta}\right) \, d\theta$. Summing up both equalities, we have $\begin{aligned}2 \int_0^1 \frac{\ln(1+x)}{1+x^2} \, dx &= \int_0^{\frac{\pi}{4}} \ln(1 + \tan \theta) \, d\theta + \int_0^{\frac{\pi}{4}} \ln\left(\frac{2}{1 + \tan \theta}\right) \, d\theta \\ &= \int_0^{\frac{\pi}{4}} \ln(1 + \tan \theta) + \ln 2 - \ln(1 + \tan \theta) \, d\theta \\ &= \int_0^{\frac{\pi}{4}} \ln 2 \, d\theta = \frac{\pi}{4} \ln 2\end{aligned}$ $\therefore \int_0^1 \frac{\ln(1+x)}{1+x^2} \, dx = \frac{\pi}{8} \ln 2$	

2 (iv)	$ \begin{aligned} \int_0^{\frac{\pi}{4}} \frac{\cos x}{\cos\left(x - \frac{\pi}{4}\right)} dx &= \int_0^{\frac{\pi}{4}} \frac{\cos\left(\frac{\pi}{4} - x\right)}{\cos\left(\left(\frac{\pi}{4} - x\right) - \frac{\pi}{4}\right)} dx \\ &= \int_0^{\frac{\pi}{4}} \frac{\cos\left(\frac{\pi}{4}\right)\cos x + \sin\left(\frac{\pi}{4}\right)\sin\left(\frac{\pi}{4}\right)}{\cos(-x)} dx \\ &= \int_0^{\frac{\pi}{4}} \frac{\cos\left(\frac{\pi}{4}\right)\cos x + \sin\left(\frac{\pi}{4}\right)\sin\left(\frac{\pi}{4}\right)}{\cos(-x)} dx \\ &= \int_0^{\frac{\pi}{4}} \frac{\frac{1}{\sqrt{2}}\cos x + \frac{1}{\sqrt{2}}\sin x}{\cos x} dx \\ &= \int_0^{\frac{\pi}{4}} \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\tan x dx \\ &= \left[\frac{x}{\sqrt{2}} + \frac{1}{\sqrt{2}}\ln \sec x \right]_0^{\frac{\pi}{4}} \\ &= \left(\frac{\pi}{4\sqrt{2}} + \frac{1}{\sqrt{2}}\ln\left \sec\left(\frac{\pi}{4}\right)\right \right) - 0 \\ &= \frac{\pi}{4\sqrt{2}} + \frac{1}{\sqrt{2}}\ln \sqrt{2} \\ &= \frac{\pi}{4\sqrt{2}} + \frac{1}{2\sqrt{2}}\ln 2 \end{aligned} $	
		Total Marks: 9

Qn	Solution	Remarks
3(a)	$\frac{1}{y+z}, \frac{1}{x+z}, \frac{1}{x+y}$ are three consecutive numbers of an AP $\Leftrightarrow \frac{1}{x+z} - \frac{1}{y+z} = \frac{1}{x+y} - \frac{1}{x+z}$ $\Leftrightarrow \frac{(y+z)-(x+z)}{(x+z)(y+z)} = \frac{(x+z)-(x+y)}{(x+y)(x+z)}$ $\Leftrightarrow \frac{y-x}{(x+z)(y+z)} = \frac{z-y}{(x+y)(x+z)}$ $\Leftrightarrow \frac{(y-x)(x+y)}{(x+y)(x+z)(y+z)} = \frac{(z-y)(y+z)}{(x+y)(x+z)(y+z)}$ $\Leftrightarrow \frac{y^2 - x^2}{(x+y)(x+z)(y+z)} = \frac{z^2 - y^2}{(x+y)(x+z)(y+z)}$ $\Leftrightarrow y^2 - x^2 = z^2 - y^2$ $\Leftrightarrow x^2, y^2, z^2 \text{ are three consecutive numbers of an AP}$	
(b) (i)	$\frac{a_{n+2} + a_n}{a_{n+1}} = \frac{\frac{a_{n+1}^2 + 2}{a_n} + a_n}{a_{n+1}}$ $= \frac{a_{n+1}^2 + 2 + a_n^2}{a_{n+1}a_n}$ $= \frac{a_{n+1} + \frac{a_n^2 + 2}{a_{n+1}}}{a_n}$ $= \frac{a_{n+1} + a_{n-1}}{a_n}, n \geq 2$	
(b) (ii)	$a_3 = \frac{a_2^2 + 2}{a_1} = \frac{1^2 + 2}{1} = 3$ $\frac{a_{n+1} + a_{n-1}}{a_n} = \dots = \frac{a_3 + a_1}{a_2} = \frac{3+1}{1} = 4$ $a_{n+1} = 4a_n - a_{n-1}, n \geq 2.$ Therefore a_{n+1} has the same parity as a_{n-1} since $4a_n$ is even. Since both a_1 and a_2 are both odd, by induction, a_n is an odd integer for all $n \in \mathbb{Z}^+$. Alternatively, let $P(n)$ be the statement that a_n is odd for $n \in \mathbb{Z}^+$. $P(1)$ and $P(2)$ are both true trivially by definition.	

	Suppose a_k and a_{k+1} are both odd for some positive integer k. Then $a_{k+2} = \frac{a_{k+1}^2 + 2}{a_k} = \frac{(\text{odd})^2 + 2}{\text{odd}} = \frac{\text{odd}}{\text{odd}} \neq \text{even}$ Hence $P(k), P(k+1)$ both true $\Rightarrow P(k+2)$ is true. Therefore, since $P(1)$ and $P(2)$ are both true, and $P(k), P(k+1)$ both true $\Rightarrow P(k+2)$ is true, by the Principle of Mathematical Induction, a_n is an odd integer for all $n \in \mathbb{Z}^+$.	
(c)	Suppose there is an arithmetic progression $a_1, a_2, \dots$ with common difference d that has $1, \sqrt{2}, \sqrt{3}$ among its terms. Then there exist distinct positive integers m, n and p such that $a_m = 1, a_n = \sqrt{2}, a_p = \sqrt{3}$. Thus we have $\sqrt{2} - 1 = a_n - a_m = (n - m)d$ and $\sqrt{3} - \sqrt{2} = a_p - a_n = (p - n)d, \text{ so } \frac{\sqrt{3} - \sqrt{2}}{\sqrt{2} - 1} = \frac{n - m}{p - n} \text{ is rational.}$ Since $\frac{\sqrt{3} - \sqrt{2}}{\sqrt{2} - 1} = (\sqrt{3} - \sqrt{2})(\sqrt{2} + 1) = \sqrt{6} - 2 + \sqrt{3} - \sqrt{2}$ thus $a = \sqrt{6} + \sqrt{3} - \sqrt{2}$ is a rational number. Then, $a + \sqrt{2} = \sqrt{6} + \sqrt{3}$ $(a + \sqrt{2})^2 = (\sqrt{6} + \sqrt{3})^2$ $a^2 + 2a\sqrt{2} + 2 = 6 + 2\sqrt{18} + 3$ $2a\sqrt{2} - 6\sqrt{2} = 7 - a^2$ $(2a - 6)\sqrt{2} = 7 - a^2$ Since $a = 3$ does not satisfy the equality, we can divide throughout by $(2a - 6)$ so $\sqrt{2} = \frac{7 - a^2}{2a - 6} \in \mathbb{Q}$ which is a contradiction. Hence the supposition does not hold and the result required is shown.	
		Total Marks: 12

Qn	Solution	Remarks
4(i)	Applying Cauchy-Schwarz Inequality $\left(\sum_{i=1}^{n+1} a_i^2\right)\left(\sum_{i=1}^{n+1} b_i^2\right) \geq \left(\sum_{i=1}^{n+1} a_i b_i\right)^2 \text{ with}$ $a_i = x_{i-1} - x_i$ for $i = 1, 2, \dots, n$, $a_{n+1} = x_n$ and $b_i = 1$ for $i = 1, 2, \dots, n+1$, we have $\left(\left(\sum_{i=1}^n (x_{i-1} - x_i)^2\right) + x_n^2\right)\left(\sum_{i=1}^{n+1} 1^2\right) \geq \left(\left(\sum_{i=1}^n (x_{i-1} - x_i)\right) + x_n\right)^2 = 1$ $\left(\left(\sum_{i=1}^n (x_{i-1} - x_i)^2\right) + x_n^2\right) \geq \frac{1}{n+1}$	
(ii)	Since equality holds, we have $a_i = mb_i = m$ (constant). Thus, $a_1 = a_2 = \dots = a_{n+1}$. Since $\sum_{i=1}^{n+1} a_i = 1$, therefore we have $a_i = \frac{1}{n+1} \text{ for } i = 1, 2, \dots, n+1,$ Thus, $x_n = \frac{1}{n+1},$ $x_{n-1} = a_n + x_n = \frac{2}{n+1},$ $x_{n-2} = a_{n-1} + x_{n-1} = \frac{3}{n+1},$ and so on so forth. Generalizing, we have $x_k = \frac{n+1-k}{n+1}, \quad k = 0, 1, 2, \dots, n.$	
		Total Marks: 6

Qn	Solution	Mark Scheme																																
5(i)	Number of ways $= \binom{24+3-1}{3-1} = 325$	B1																																
(ii)	The problem is $x_1 + x_2 + x_3 = 24$ $0 \leq x_i \leq 10, i = 1, 2, 3$ with x_1, x_2, x_3 being the number of cards in the red, green and blue boxes respectively. Method 1 Let A_i denote the event where $0 \leq x_i \leq 10$. Number of ways $ A_1 \cap A_2 \cap A_3 $ $= S - A'_1 \cup A'_2 \cup A'_3 $ $= S - \left(\sum_{i=1}^3 A'_i - \sum_{\substack{i \neq j \\ i, j \in \{1, 2, 3\}}} A'_i \cap A'_j + A'_1 \cap A'_2 \cap A'_3 \right)$ $= 325 - \left(3 \binom{13+3-1}{3-1} - 3 \binom{2+3-1}{3-1} + 0 \right)$ $= 325 - (3(105) - 3(6) + 0) = 28$ Method 2 $x_i \geq 4$ since $x_i \leq 3$ for any i will result in $x_1 + x_2 + x_3 < 24$ for $0 \leq x_i \leq 10, i = 1, 2, 3$. <table><tr><th>$x_1$</th><th>$x_2 + x_3$</th><th>$(x_2, x_3)$ or corresponding cases for x_2</th><th>Number of Ways</th></tr><tr><td>4</td><td>20</td><td>(10, 10)</td><td>1</td></tr><tr><td>5</td><td>19</td><td>(9, 10), (10, 9)</td><td>2</td></tr><tr><td>6</td><td>18</td><td>$8 \leq x_2 \leq 10$</td><td>3</td></tr><tr><td>7</td><td>17</td><td>$7 \leq x_2 \leq 10$</td><td>4</td></tr><tr><td>8</td><td>16</td><td>$6 \leq x_2 \leq 10$</td><td>5</td></tr><tr><td>9</td><td>15</td><td>$5 \leq x_2 \leq 10$</td><td>6</td></tr><tr><td>10</td><td>14</td><td>$4 \leq x_2 \leq 10$</td><td>7</td></tr></table> Therefore number of ways $= 1 + 2 + \dots + 7 = 28$	x_1	$x_2 + x_3$	(x_2, x_3) or corresponding cases for x_2	Number of Ways	4	20	(10, 10)	1	5	19	(9, 10), (10, 9)	2	6	18	$8 \leq x_2 \leq 10$	3	7	17	$7 \leq x_2 \leq 10$	4	8	16	$6 \leq x_2 \leq 10$	5	9	15	$5 \leq x_2 \leq 10$	6	10	14	$4 \leq x_2 \leq 10$	7	
x_1	$x_2 + x_3$	(x_2, x_3) or corresponding cases for x_2	Number of Ways																															
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10	14	$4 \leq x_2 \leq 10$	7																															

(ii)	Method 3 Number of ways $= (\text{none with 10 units}) + (\text{one with 10 units})$ $+ (\text{two with 10 units})$ $= \left(\begin{matrix} 1 & + & 3 & + & 6 \\ (8,8,8) & (6,9,9)^* & (7,8,9)^* \end{matrix} \right) + \underbrace{5}_{(5,9) \dots (9,5)} \times {}^3C_1 + {}^3C_2_{(4,10,10)^*}$ $= 10 + 15 + 3 = 28$	
(iii)	Sue can take either no work, or up to 4 units of work, since it is not possible to have Sue take 5 or more units of work as the total amount of work is 24 units ($5 + 10 + 10 = 25 > 24$). If Sue takes k units of work where $k = 0$ to 4, then the other 2 must take at least $2k$ units of work from the remaining $(24 - k)$ units of work, no. of ways $= \binom{24 - k - 4k + 2 - 1}{2 - 1} = 25 - 5k$ since k units of work is assigned to Sue, and we place $2k$ units of work each into the other two boxes. Number of ways supervisor can distribute the workload $= \sum_{k=0}^4 (25 - 5k) = 75$	
(iv)	$P(r, n) = P(r - 1, n - 1) + P(r - n, n)$	
(v)	$\begin{aligned} &P(r, n) \\ &= P(r - 1, n - 1) + P(r - n, n) \\ &(\text{applying (iv) on } P(r, n)) \\ &= [P(r - 2, n - 2) + P((r - 1) - (n - 1), (n - 1) - 1)] \\ &\quad + P(r - n, n) \\ &(\text{applying (iv) on } P(r - 1, n - 1)) \\ &= P(r - 2, n - 2) + [P(r - n, n - 1) + P(r - n, n)] \\ &(\text{rearrangement}) \\ &= P(r - (n - 1), 1) \\ &\quad + [P(r - n, n - (n - 2)) + \dots + P(r - n, n)] \\ &(\text{applying (iv) on } P(r - 2, n - 2), \dots, \text{ until } P(r - n + 1, 1)) \\ &= P(r - n + 1, 1) + [P(r - n, 2) + \dots + P(r - n, n)] \\ &= P(r - n, 1) + [P(r - n, 2) + \dots + P(r - n, n)] \\ &(\because P(r - n + 1, 1) = P(r - n, 1) = 1) \\ &= \sum_{k=1}^n P(r - n, k) \quad (\text{shown}) \end{aligned}$	

(vi)	<u>Method 1 (Using (v))</u> $ \begin{aligned} P(10, 3) &= \sum_{k=1}^3 P(7, k) \\ &= P(7, 1) + P(7, 2) + P(7, 3) \\ &= 1 + \sum_{k=1}^2 P(5, k) + \sum_{k=1}^3 P(4, k) \\ &= 1 + P(5, 1) + P(5, 2) + P(4, 1) + P(4, 2) + P(4, 3) \\ &= 1 + 1 + P(5, 2) + 1 + P(4, 2) + 1 \\ &= 4 + \sum_{k=1}^2 P(3, k) + \sum_{k=1}^2 P(2, k) \\ &= 4 + P(3, 1) + P(3, 2) + P(2, 1) + P(2, 2) \\ &= 4 + 1 + 1 + 1 + 1 = 8 \end{aligned} $ <u>Method 2 (Using (iv))</u> $ \begin{aligned} P(10, 3) &= P(9, 2) + P(7, 3) \\ &= (P(8, 1) + P(7, 2)) + (P(6, 2) + P(4, 3)) \\ &= 1 + (P(6, 1) + P(5, 2)) + (P(5, 1) + P(4, 2)) + 1 \\ &= 1 + 1 + (P(4, 1) + P(3, 2)) + 1 + (P(3, 1) + P(2, 2)) + 1 \\ &= 8 \end{aligned} $	
		Total Marks: 12

Qn	Solution	Remarks
6(i) (a)	$T_1 = 1, T_2 = 2.$	
(i) (b)	$T_{n+2} = T_{n+1} + T_n$ for $n \geq 1.$	
(ii)	Consider the “odd” tile / last tile in a tiling of P_{n+1}. It can only be covered by a 1×1 or a 1×2 tile. If it is covered by a 1×1 tile, the rest form a tiling of Q_n. If it is covered by a 1×2 tile, the rest form a tiling of P_n. Thus $P_{n+1} = P_n + Q_n$.	
(iii)	Consider the last column of 2 tiles in a tiling of Q_{n+1}. The following cases are possible: • 2×2 tile: The rest form a tiling of Q_{n-1}. • 1×2 tile (vertical): The rest form a tiling of Q_n • Two 1×1 tiles: The rest form a tiling of Q_n. • Two 1×2 tiles (horizontal): The rest form a tiling of Q_{n-1}. • One 1×1 tile and one 1×2 tile (horizontal): The rest form a tiling of P_n. Note that this case counts twice (depending on which tile covers the top line and which tile covers the bottom line). Thus $Q_{n+1} = 2Q_n + 2Q_{n-1} + 2P_n$ $= 2P_{n+1} + 2Q_{n-1}$ using the result from (ii). <u>Alternative</u> Tiling of Q_{n+1} can be obtained from • Q_{n-1} using a 2×2 tile or two horizontal 1×2 tiles. • Q_n using one vertical 1×2 tile • P_{n+1} using one 1×1 tile The first two cases involve non-1×1 tiles on the last column, and the last case involves 1×1 tiles on the last column. However, the last case we must consider the double counting of the subcase where tiles from the last column are both 1×1 tiles, and the number of ways to do so is Q_n. Thus $Q_{n+1} = 2Q_{n-1} + Q_n + (2P_{n+1} - Q_n) = 2P_{n+1} + 2Q_{n-1}$	

(iv)	Add $P_{n+1} + 2P_{n-1}$ to both sides of (iii): $P_{n+1} + 2P_{n-1} + Q_{n+1} = P_{n+1} + 2P_{n-1} + 2P_{n+1} + 2Q_{n-1}$ $P_{n+2} + 2P_{n-1} = 3P_{n+1} + 2P_n \quad (\text{using result from (ii)})$	
(v)	Number of tilings of $2 \times n$ path is Q_n So $Q_n = P_{n+1} - P_n$ $= \frac{2}{7}(-1)^n + \frac{1+2\sqrt{2}}{14}(2+\sqrt{2})^n(2+\sqrt{2}-1)$ $+ \frac{1-2\sqrt{2}}{14}(2-\sqrt{2})^n(2-\sqrt{2}-1)$ $= \frac{2}{7}(-1)^n + \frac{5+3\sqrt{2}}{14}(2+\sqrt{2})^n + \frac{5-3\sqrt{2}}{14}(2-\sqrt{2})^n$	
		Total Marks: 12

Qn	Solution	Remarks														
7(a)	We show by proving the contrapositive statement: if $7 \nmid a$ or $7 \nmid b$ then $7 \nmid (a^2 + b^2)$. For positive integers n not divisible by 7, <table><tr><th>$n \pmod{7}$</th><th>$n^2 \pmod{7}$</th></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>4</td></tr><tr><td>3</td><td>2</td></tr><tr><td>4</td><td>2</td></tr><tr><td>5</td><td>4</td></tr><tr><td>6</td><td>1</td></tr></table> Therefore $(a^2 + b^2) \pmod{7}$ can only take values 1 (from $1 + 1$), 2 (from $1 + 2$), 3 (from $1 + 2$), 4 (from $2 + 2$), 5 (from $1 + 4$) 6 (from $2 + 4$) but never 0. Hence the contrapositive statement is shown.	$n \pmod{7}$	$n^2 \pmod{7}$	1	1	2	4	3	2	4	2	5	4	6	1	
$n \pmod{7}$	$n^2 \pmod{7}$															
1	1															
2	4															
3	2															
4	2															
5	4															
6	1															
(b)	When n is even, $n^4 + 4^n$ is even and hence not a prime. When n is odd, i.e. $n = 2k + 1$ with integer $k \geq 1$, (since $n > 1$) $\begin{aligned} n^4 + 4^n &= (n^2 + 2^n)^2 - 2(n^2)(2^n) \\ &= (n^2 + 2^n)^2 - (n^2)(2^{2k+2}) \\ &= (n^2 + 2^n)^2 - (2^{k+1}n)^2 \\ &= (n^2 + 2^n - 2^{k+1}n)(n^2 + 2^n + 2^{k+1}n) \end{aligned}$ The smaller factor is $(n^2 + 2^n - 2^{k+1}n)$, and $\begin{aligned} n^2 + 2^n - 2^{k+1}n &= n^2 - 2^{k+1}n + 2^n \\ &= n^2 - 2(2^k)n + 2^{2k} - 2^{2k} + 2^{2k+1} \\ &= (n - 2^k)^2 + 2^{2k} > 1 \end{aligned}$ so $n^4 + 4^n$ cannot be prime.															

7(c)	<u>Method 1: Consider the values of</u> 16^n, $10n$ and $-1 \pmod{25}$ <table><tr><th>n</th><th>$16^n \pmod{25}$</th><th>$10n \pmod{25}$</th><th>-1</th><th>Sum $\pmod{25}$</th></tr><tr><td>1</td><td>16</td><td>10</td><td>-1</td><td>0</td></tr><tr><td>2</td><td>6</td><td>20</td><td>-1</td><td>0</td></tr><tr><td>3</td><td>21</td><td>5</td><td>-1</td><td>0</td></tr><tr><td>4</td><td>11</td><td>15</td><td>-1</td><td>0</td></tr><tr><td>5</td><td>1</td><td>0</td><td>-1</td><td>0</td></tr><tr><td>6</td><td>16</td><td>10</td><td>-1</td><td>0</td></tr><tr><td>7</td><td>6</td><td>20</td><td>-1</td><td>0</td></tr><tr><td>$\vdots$</td><td>$\vdots$</td><td>$\vdots$</td><td>$\vdots$</td><td>$\vdots$</td></tr></table> Since the table repeats cyclically for every 5 values of n, $16^n + 10n - 1$ is divisible by 25, for $n \in \mathbb{Z}^+$. <u>Method 2: Mathematical Induction</u> Let $P(n)$ be the statement that $16^n + 10n - 1$ is divisible by 25, for $n \in \mathbb{Z}^+$. $P(1)$ is true since $16^1 + 10(1) - 1 = 25$ which is divisible by 25. Suppose $P(k)$ is true for some positive integer k. Then for $P(k + 1)$, $16^{k+1} + 10(k + 1) - 1$ $= 16((16^k) + 10k - 1) - 160k + 16 + 10(k + 1) - 1$ $= 16((16^k) + 10k - 1) - 150k + 25$ which is divisible by 25. Hence $P(k)$ is true $\Rightarrow P(k + 1)$ is true. Therefore, since $P(1)$ is true, and $P(k)$ is true $\Rightarrow P(k + 1)$ is true, by the Principle of Mathematical Induction, $16^n + 10n - 1$ is divisible by 25, for $n \in \mathbb{Z}^+$.	n	$16^n \pmod{25}$	$10n \pmod{25}$	-1	Sum $\pmod{25}$	1	16	10	-1	0	2	6	20	-1	0	3	21	5	-1	0	4	11	15	-1	0	5	1	0	-1	0	6	16	10	-1	0	7	6	20	-1	0	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	
n	$16^n \pmod{25}$	$10n \pmod{25}$	-1	Sum $\pmod{25}$																																											
1	16	10	-1	0																																											
2	6	20	-1	0																																											
3	21	5	-1	0																																											
4	11	15	-1	0																																											
5	1	0	-1	0																																											
6	16	10	-1	0																																											
7	6	20	-1	0																																											
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$																																											
		Total Marks: 12																																													

Qn	Solution	Remarks
8(i)	$L_1(x_1) = 1,$ $L_1(x_2) = 0, L_1(x_3) = 0.$	
(ii)	$L_2(x) = \frac{(x-x_1)(x-x_3)}{(x_2-x_1)(x_2-x_3)}$ $L_3(x) = \frac{(x-x_1)(x-x_2)}{(x_3-x_1)(x_3-x_2)}$	
(iii)	Suppose there exist two quadratic polynomials p_1, p_2 (of degree less or equal to $n-1=3-1=2$), with $p_1(x_i) = p_2(x_i) = y_i$, for $i=1, 2, 3$. Then the difference polynomial $q = p_1 - p_2$ is a polynomial of degree less or equal to $n-1=3-1=2$ that satisfy $h(x_i) = 0$, for $i=1, 2, 3$. Since the number of zeroes of a nonzero polynomial is equal to its degree, it follows that $q(x) = p_1(x) - p_2(x) = 0, x \in \mathbb{R}$, i.e. $p_1(x) = p_2(x)$, for all $x \in \mathbb{R}$.	
(iv)	$P(x) = y_1 L_1(x) + y_2 L_2(x) + y_3 L_3(x), x \in \mathbb{R}.$ This is because since $L_i(x_i) = 1, L_i(x_j) = 0$ for $i \neq j$, $i, j = 1, 2, 3$ from the earlier parts, we have $P(x_i) = y_i$, for $i=1, 2, 3$.	
(v)	Let $X = Y = \{1, 2, 3\}$ and a function $f : X \rightarrow X$, satisfying $f(x_i) = y_i, y_i \neq i$ for $i=1, 2, 3$, where $x_i = i$, and $y_i \in Y = X$, for $i=1, 2, 3$. Note that the above translates to a derangement problem, where the number of such possible mappings is given by $D_3 = 3! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} \right) = 2$. In fact, the 2 derangements are given by Derangement 1: $f(1) = 2, f(2) = 3, f(3) = 1$, Derangement 2: $f(1) = 3, f(2) = 1, f(3) = 2$. We now show that we can extend this derangement property of function f defined on $\{1, 2, 3\}$ to a quadratic polynomial Q, which is defined on the real line by finding explicitly 2 quadratic polynomials Q_1, Q_2 such that they satisfy $Q_1(1) = 2, Q_1(2) = 3, Q_1(3) = 1$, $Q_2(1) = 3, Q_2(2) = 1, Q_2(3) = 2$.	

To do so, we can use either of the following two methods: Method 1 We can find two quadratic polynomials Q_1, Q_2 such that they satisfy $Q_1(1) = 2, Q_1(2) = 3, Q_1(3) = 1,$ $Q_2(1) = 3, Q_2(2) = 1, Q_2(3) = 2.$ from parts (iii) and the interpolatory property (iv), we get $Q_1(x) = 2L_1(x) + 3L_2(x) + 1L_3(x),$ $Q_2(x) = 3L_1(x) + 1L_2(x) + 2L_3(x),$ where $L_1(x) = \frac{(x-2)(x-3)}{(1-2)(1-3)} = \frac{1}{2}(x-2)(x-3),$ $L_2(x) = \frac{(x-1)(x-3)}{(2-1)(2-3)} = -(x-1)(x-3),$ $L_3(x) = \frac{(x-1)(x-2)}{(3-1)(3-2)} = \frac{1}{2}(x-1)(x-2).$ Method 2 Let $Q_1(x) = a_2x^2 + a_1x + a_0$, and $Q_2(x) = b_2x^2 + b_1x + b_0.$ We solve for coefficients a_2, a_1, a_0 and b_2, b_1, b_0 such that $Q_1(1) = 2, Q_1(2) = 3, Q_1(3) = 1,$ or $Q_2(1) = 3, Q_2(2) = 1, Q_2(3) = 2.$ We obtain the resulting system of linear equations $a_2(1)^2 + a_1(1) + a_0 = 2,$ $a_2(2)^2 + a_1(2) + a_0 = 3,$ $a_2(3)^2 + a_1(3) + a_0 = 1.$ $b_2(1)^2 + b_1(1) + b_0 = 3,$ $b_2(2)^2 + b_1(2) + b_0 = 1,$ $b_2(3)^2 + b_1(3) + b_0 = 2.$ $a_2 = -\frac{3}{2}, a_1 = \frac{11}{2}, a_0 = -2,$ $b_2 = \frac{3}{2}, b_1 = -\frac{13}{2}, b_0 = 8,$ Therefore, $Q_1(x) = -\frac{3}{2}x^2 + \frac{11}{2}x - 2,$ $Q_2(x) = \frac{3}{2}x^2 - \frac{13}{2}x + 8.$	Total Marks: 13
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Qn	Solution	Remarks
9(i) (a)	$S(p) = p + 1$	
(i) (b)	$S(pq) = S(p)S(q) = (p+1)(q+1)$	
(i) (c)	$S(p^m q^n) = S(p^m)S(q^n)$ $= (1 + p + \dots + p^m)(1 + q + \dots + q^n)$ $= \frac{(p^{m+1} - 1)(q^{n+1} - 1)}{(p - 1)(q - 1)}$	
(ii)	$220 = 2^2 \times 5 \times 11$ The proper divisors of 220 are 1, 2, 4, 5, 10, 11, 20, 22, 44, 55, and 110, and Sum of all proper divisors of 220 $= 1 + 2 + 4 + 5 + 10 + 11 + 20 + 22 + 44 + 55 + 110$ $= 284$ $284 = 2^2 \times 71$ The proper divisors of 284 are 1, 2, 4, 71 and 142, and Sum of all proper divisors of 284 $= 1 + 2 + 4 + 71 + 142 = 220$	
(iv)	$a(pq + r) = M + N$ $= S(M)$ $= S(apq)$ $= S(a)S(p)S(q)$ $= (S(a))(p+1)(q+1)$	
(v)	Since M , N form a pair of amicable numbers, $S(M) - M = N$, and $S(N) - N = M$. Simplifying, we get $S(M) = S(N)$, and thus $S(apq) = S(ar)$ $(p+1)(q+1)S(a) = (r+1)S(a)$ $r+1 = (p+1)(q+1)$ $(2a - S(a))(p+1)(q+1)$ $= 2a(p+1)(q+1) - S(a)(p+1)(q+1)$ $= 2a(p+1)(q+1) - a(pq+r)$ $= 2a(p+1)(q+1) - a(pq + (p+1)(q+1) - 1)$ $= 2a(pq + p + q + 1) - a(2pq + p + q)$ $= (2apq + 2ap + 2aq + 2a) - (2apq + ap + aq)$ $= (ap + aq + 2a) = a(p + q + 2) \text{ (shown)}$	

(vi)	$a = 4, S(4) = 1 + 2 + 4 = 7.$ $\left(p+1-\frac{a}{2a-S(a)}\right)\left(q+1-\frac{a}{2a-S(a)}\right)=\left(\frac{a}{2a-S(a)}\right)^2$ becomes $\left(p+1-\frac{4}{2(4)-7}\right)\left(q+1-\frac{4}{2(4)-7}\right)=\left(\frac{4}{2(4)-7}\right)^2$ $(p-3)(q-3)=16$ WLOG let $p \leq q,$ <table><tr><th>$p-3$</th><th>$q-3$</th><th>p</th><th>q</th><th>r</th></tr><tr><td>1</td><td>16</td><td>4</td><td>19</td><td>99</td></tr><tr><td>2</td><td>8</td><td>5</td><td>11</td><td>71</td></tr><tr><td>4</td><td>4</td><td>7</td><td>7</td><td>63</td></tr></table> Since p, q, r are distinct primes, so the only solution is $p = 5, q = 11, r = 71,$ and $M = 4 \times 5 \times 11 = 220, N = 4 \times 71 = 284.$	$p-3$	$q-3$	p	q	r	1	16	4	19	99	2	8	5	11	71	4	4	7	7	63	
$p-3$	$q-3$	p	q	r																		
1	16	4	19	99																		
2	8	5	11	71																		
4	4	7	7	63																		
		Total Marks: 13																				